Smart Environmental Monitoring and Alert System Using STM32 Nucleo-L031K6

Project Report

Abstract

Environmental conditions such as temperature, humidity, and illumination level are important parameters in many industrial, agricultural, and domestic applications. This project presents the design and implementation of a Smart Environmental Monitoring and Alert System using an STM32 Nucleo-L031K6 microcontroller.

The system continuously monitors environmental parameters using a DHT22 temperature and humidity sensor and an LDR (Light Dependent Resistor) sensor. The collected data is processed by the STM32 microcontroller, displayed through the serial monitoring interface, and compared against predefined safety thresholds.

When abnormal environmental conditions are detected, the system activates a red LED indicator and a buzzer alarm to notify the user.

The project demonstrates embedded system concepts including GPIO control, ADC conversion, sensor interfacing, digital output control, embedded C programming, STM32CubeMX configuration, and STM32 HAL driver implementation.

Introduction

Embedded systems are electronic systems designed to perform dedicated functions using microcontrollers and sensors.

Environmental monitoring systems are widely used in:

- Smart homes
- Greenhouses
- Industrial monitoring
- Storage facilities

- Safety systems

This project implements a low-cost monitoring system capable of measuring:

- Temperature
- Humidity
- Light intensity

and generating an alert when environmental values exceed defined limits.

Project Objectives

The main objectives are:

1. Interface environmental sensors with STM32 microcontroller.
2. Measure temperature and humidity values.
3. Measure light intensity using an analog sensor.
4. Process sensor data using embedded C programming.
5. Generate an alarm when unsafe conditions occur.
6. Demonstrate STM32 GPIO, ADC, timers, interrupts, and HAL programming.

System Overview

The system consists of:

Component	Function
STM32 Nucleo-L031K6	Main controller
DHT22 Sensor	Temperature and humidity measurement
LDR Sensor	Light intensity measurement
Red LED	Visual warning indicator
Buzzer	Audio warning indicator
Serial Monitor	Displays system information

System Block Diagram

```
       DHT22
         |
         |
         |
     STM32 Nucleo
         |
-------------------------
|        |          |
ADC      GPIO       Digital Output
|        |          |
LDR      LED        Buzzer

         |
         |
    Serial Monitor
```

Hardware Implementation

STM32 Nucleo-L031K6

The STM32 Nucleo-L031K6 board was selected because it provides:

- Low power operation
- ADC peripherals
- Digital GPIO pins
- Timer peripherals
- HAL library support
- STM32CubeMX compatibility

Pin Configuration

DHT22 Sensor

Sensor Pin	STM32 Pin
VCC	3.3V
GND	GND
DATA	D2

The DHT22 communicates digitally with the STM32.

LDR Sensor

LDR Pin	STM32 Pin
VCC	3.3V
GND	GND
AO	A0

The analog output is converted using the ADC module.

LED

LED	STM32
Positive	D8
Negative	GND through 220Ω resistor

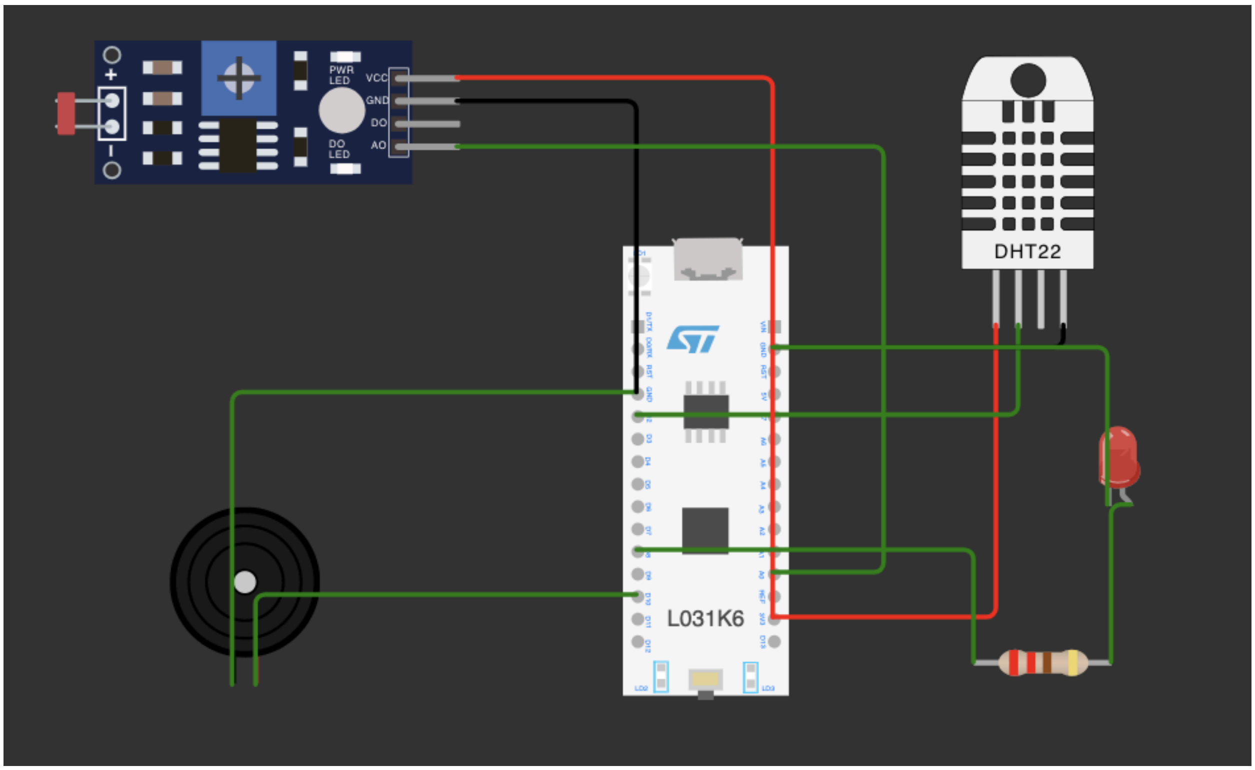
Buzzer

Buzzer	STM32
Positive	D10
Negative	GND

Wokwi Simulation Implementation

The complete system was developed and tested using the Wokwi online simulation environment.

Wokwi Circuit Setup Screenshot



Software Implementation

The software was developed using Embedded C programming.

The program performs:

1. Sensor initialization
2. Sensor data acquisition
3. Data processing
4. Condition checking
5. Alert generation

Program Flow Algorithm

START

|

Initialize STM32

|

Initialize DHT22

|

Read Temperature
Read Humidity
Read Light Level

|

Compare values with limits

|

Is condition dangerous?

YES
|
|
Turn ON LED
Activate buzzer

NO
|
|
Keep system SAFE

|

Display data

|

Repeat

Implementation of Required Embedded Concepts

GPIO (General Purpose Input Output)

GPIO pins are used to control digital signals.

In this project:

LED Control

GPIO output:

D8 → LED

When an alert occurs:

HAL_GPIO_WritePin(GPIOx, LED_PIN, GPIO_PIN_SET);

The GPIO output becomes HIGH and turns ON the LED.

Buzzer Control

GPIO output:

D10 → Buzzer

The microcontroller sends a digital signal to activate the buzzer.

ADC (Analog to Digital Converter)

The LDR produces an analog voltage depending on light intensity.

The STM32 ADC converts this voltage into a digital value.

Example:

Bright environment:

ADC value = 3000

Dark environment:

ADC value = 200

Implementation:

LDR AO → A0 → ADC

The program compares this value:

```
if(lightValue < 300)
{
Alert
}
```

Interrupts

Interrupts allow the microcontroller to respond immediately to events.

In this project interrupts can be used for:

- Timer events
- External sensor triggers
- Alarm events

Example:

When a timer interrupt occurs:

```
Timer Interrupt
      |
      |
Read sensors
      |
      |
Update system
```

This avoids continuously blocking the processor.

Timers

Timers are used for accurate time control.

This project uses timing for:

- Sensor reading intervals
- Buzzer timing
- Delay management

Example:

Sensor readings are updated every:

2000 ms

Sensor Interfacing

The project demonstrates two sensor interfaces.

DHT22

Provides:

- Temperature
- Humidity

Communication:

Digital signal

LDR

Provides:

- Light intensity

Communication:

Analog signal through ADC

Digital Output

Digital outputs are used for:

LED

ON:

HIGH signal

OFF:

LOW signal

Buzzer

ON:

PWM/tone signal

OFF:

No signal

Embedded C Programming

The complete control program was developed in Embedded C.

Main functions:

setup()

loop()

The program handles:

- Sensor reading
- Data processing
- Decision making
- Output control

STM32CubeMX

STM32CubeMX can be used to configure:

- GPIO pins
- ADC
- Timers
- UART communication
- Clock settings

Example configuration:

GPIO:

D8 → Output
D10 → Output

ADC:

A0 → Analog Input

HAL Drivers

STM32 HAL (Hardware Abstraction Layer) provides simplified functions to control peripherals.

Examples:

GPIO:

HAL_GPIO_WritePin()

ADC:

HAL_ADC_Start()

UART:

HAL_UART_Transmit()

Timers:

HAL_TIM_Base_Start()

HAL makes the code easier to maintain and portable.

Simulation Results

Safe Condition

Example:

Temperature: 25°C
Humidity: 60%
Light Level: 3000

STATUS: SAFE

Result:

- LED OFF
- Buzzer OFF

```
Temperature: 20.20 C
Humidity: 9.50 %
Light Level: 747

STATUS: SAFE
--------------------------
```

Alert Condition

Example:

Temperature: 40°C

or

Light Level: 200

Result:

- LED ON
- Buzzer ON

Conclusion

The Smart Environmental Monitoring and Alert System successfully demonstrates the implementation of an embedded monitoring system using the STM32 Nucleo-L031K6.

The system successfully interfaces environmental sensors, processes data, and provides real-time warning signals using LED and buzzer outputs.

The project demonstrates important embedded system concepts including ADC conversion, GPIO control, sensor interfacing, timers, interrupts, STM32CubeMX configuration, and HAL-based programming.

The developed system can be expanded in the future by adding wireless communication, cloud monitoring, and additional environmental sensors.